CSA1708-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENTS

Program 14: WRITE THE PYTHON POGRAM TO IMPLEMENT APHA & BETA PRUNING ALGORITHM FOR GAMING

AIM

To write a Python program to implement the **Alpha-Beta Pruning algorithm** for gaming to find the optimal move by eliminating unnecessary branches in the game tree.

ALGORITHM

1. Start the program.
2. Define the game tree with leaf node values.
3. Initialize alpha to negative infinity and beta to positive infinity.
4. For the maximizing player, select the maximum value and update alpha.
5. For the minimizing player, select the minimum value and update beta.
6. If alpha is greater than or equal to beta, prune the remaining branches.
7. Continue until the optimal value is obtained.
8. Display the optimal value.
9. Stop the program.

PY8.py - C:/Users/navit/PY8.py (3.12.10)

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```
                node * 2 + i,
                False,
                values,
                alpha,
                beta
            )

            best = max(best, value)
            alpha = max(alpha, best)

            if beta <= alpha:
                print("Branch pruned")
                break

        return best

    else:
        best = float('inf')

        for i in range(2):
            value = alpha_beta(
                depth + 1,
                node * 2 + i,
                True,
                values,
                alpha,
                beta
            )

            best = min(best, value)
            beta = min(beta, best)

            if beta <= alpha:
                print("Branch pruned")
                break

        return best


values = [3, 5, 6, 9, 1, 2, 0, -1]

result = alpha_beta(
    0, 0, True, values,
    float('-inf'), float('inf')
)

print("Optimal Value:", result)
```

IDLE Shell 3.12.10

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```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/navit/PY8.py ==========
Branch pruned
Branch pruned
Optimal Value: 5
>>>
```

RESULT

The Python program was executed successfully, and the Alpha-Beta Pruning algorithm found the optimal value while pruning unnecessary branches